

CareerGhana Python Internship — Task 3

Deliverable: Career Path Predictor (First Machine Learning Model) & Writeup
Repository: https://github.com/J-git22/careerghana_tasks/tree/main/task_3_career_predictor

1. What Problem Does This Model Solve?

When aspiring tech professionals enter the field, they often wonder which career track best fits their background and personal aptitude. This project develops a supervised machine learning model using Python, Pandas, and Scikit-Learn that evaluates 5 technical skill dimensions and recommends the most aligned tech career track:

- Software Engineering
- Data Science
- Cybersecurity
- UI/UX Design

2. The Dataset Architecture (dataset.csv)

The model learns patterns from a curated dataset of 45 student/professional skill profiles rated across 5 core technical dimensions (scale 1 to 5):

Feature Name	Skill Domain Evaluated	Scale
coding	Programming syntax, algorithms, data structures & OOP	1 - 5
math_statistics	Probability, linear algebra, statistical thinking	1 - 5
networking_security	Protocols, system vulnerabilities, defensive security	1 - 5
visual_design	UI wireframing, design principles, usability testing	1 - 5
cloud_sysadmin	Linux shell, cloud architecture, CI/CD, containers	1 - 5

3. How the Model Works: Decision Tree Algorithm

We selected a **Decision Tree Classifier** because of its transparency, explainability, and speed on structured feature sets. It operates like an automated game of *20 Questions* by recursively identifying optimal thresholds that partition candidates into homogeneous career groups based on Gini impurity minimization.

4. Training, Evaluation & Accuracy

The dataset was divided using a **stratified 75% train / 25% test split** to validate that the model generalizes to unseen candidates rather than simply memorizing training profiles.

Metric / Feature	Result / Contribution
Model Test Accuracy	91.7% on holdout test set
Top Feature 1: math_statistics	35.5% feature importance (primary discriminator for Data Science)
Top Feature 2: coding	34.3% feature importance (differentiates Software Engineering)
Top Feature 3: networking_security	30.3% feature importance (primary driver for Cybersecurity)

5. Verification & Execution Output

```
> python career_predictor.py --coding 5 --math 2 --security 1 --design 1 --cloud 3
Input Profile: {'coding': 5, 'math_statistics': 2, 'networking_security': 1, 'visual_design': 1,
'cloud_sysadmin': 3}
Predicted Career: >>> SOFTWARE ENGINEERING <<<
```

Confidence: Software Engineering: 100.0% | Cybersecurity: 0.0% | Data Science: 0.0% | UI/UX: 0.0%

6. Summary & Recommendations

The model successfully delivers clear, interpretable, data-driven career recommendations. Its high explainability ensures users understand exactly why a pathway is recommended based on their technical aptitudes.